

TUMISO KOPANO NGWAKO

Cybersecurity Intern · Web Application Security & SOC · OT/ICS Security

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PROFESSIONAL SUMMARY

Final-year Computer Systems Engineering student currently on Work-Integrated Learning (WIL) as a Cybersecurity Intern, gaining hands-on exposure to OT/ICS security on a critical-infrastructure SCADA modernisation project. Practical experience across web application penetration testing (OWASP Top 10, Burp Suite), SOC/SIEM operations (Wazuh, Microsoft Sentinel), and industrial control system security (Siemens S7 PLCs). Strong engineering foundation in embedded systems, PLC/ladder logic, and low-level programming, bridging IT and OT security disciplines.

CERTIFICATIONS & TRAINING

- MWR CyberSec Virtual Internship Certificate — Web application penetration testing methodology
- EC-Council SOC Analyst (CSA) Short Course — SOC procedures, threat monitoring, incident response
- Sigma White Belt Certification — Detection engineering and Sigma rule fundamentals
- Blue Teaming Path (Modules 1–3) — Defensive security, threat hunting, incident response
- SOC Analyst Training (Tier 1, 2 & 3) — Alert triage, investigation, escalation procedures

TECHNICAL SKILLS

Web Application Security & SOC

- Burp Suite · SQL Injection, XSS, CSRF, LFI · Nmap & Reconnaissance · Kali Linux
- Wazuh, Microsoft Sentinel (SIEM) · EDR Tooling · Azure Cloud · VMware

OT / ICS Security

- Siemens S7-1200 PLC exploitation · python-snap7 · SCADA architecture
- Wireshark (industrial protocol analysis) · network segmentation · ICS exploitation techniques

Programming & Scripting

- C++, C#, Python, Assembly, SQL, Ladder Logic

Other

- Circuit design & assembly, soldering, electronic prototyping, schematic reading, troubleshooting · Microsoft Office

CYBERSECURITY EXPERIENCE & PROJECTS

Cybersecurity Intern — Work-Integrated Learning (WIL)

Current

- Hands-on exposure to Cybersecurity fundamentals, cloud security, penetration testing, ,SOC Analyst, and OT/ICS security.

MWR CyberSec Internship: Web Application Penetration Testing

2026

- Structured web application penetration testing internship covering OWASP Top 10 vulnerabilities (SQLi, XSS, CSRF, and more) and use of tools such as Burp Suite within a VMware lab environment.

Cloud Honeypot: Azure, Microsoft Sentinel

2025

- Deployed an Azure-based honeypot to capture and analyse real-world attack traffic, integrated with Microsoft Sentinel for SIEM correlation and EDR tooling.

SOC Automation Homelab: Wazuh

2025

- Built a home SOC lab using Wazuh for real-time Windows endpoint monitoring, log aggregation, and automated alerting in a virtualised environment.

OT Security: PLC Exploitation Lab

Ongoing

- Home lab built around a Siemens S7-1200 (CPU 1211C). Custom scripts and Kali Linux tooling used to identify and exploit vulnerabilities in industrial control communications.

FOUNDATIONAL ENGINEERING PROJECTS

Conveyor Belt Sorting System (PLC / Ladder Logic)

- Designed and built a conveyor belt system that separated metal from plastic objects using a PLC programmed in Ladder Logic — direct precursor to current Siemens S7 OT/ICS security work.

EDUCATION

Computer Systems Engineering

2022 – 2026

Tshwane University of Technology, Soshanguve South Campus, Gauteng — Final year, currently completing Work-Integrated Learning (WIL) as a Cybersecurity Intern.

Secondary School

2019

Moreri Secondary School, Tseoge, North West

LANGUAGES

Setswana (native) · English (fluent)